

Part A — Paràmetres de normalització (transforms)

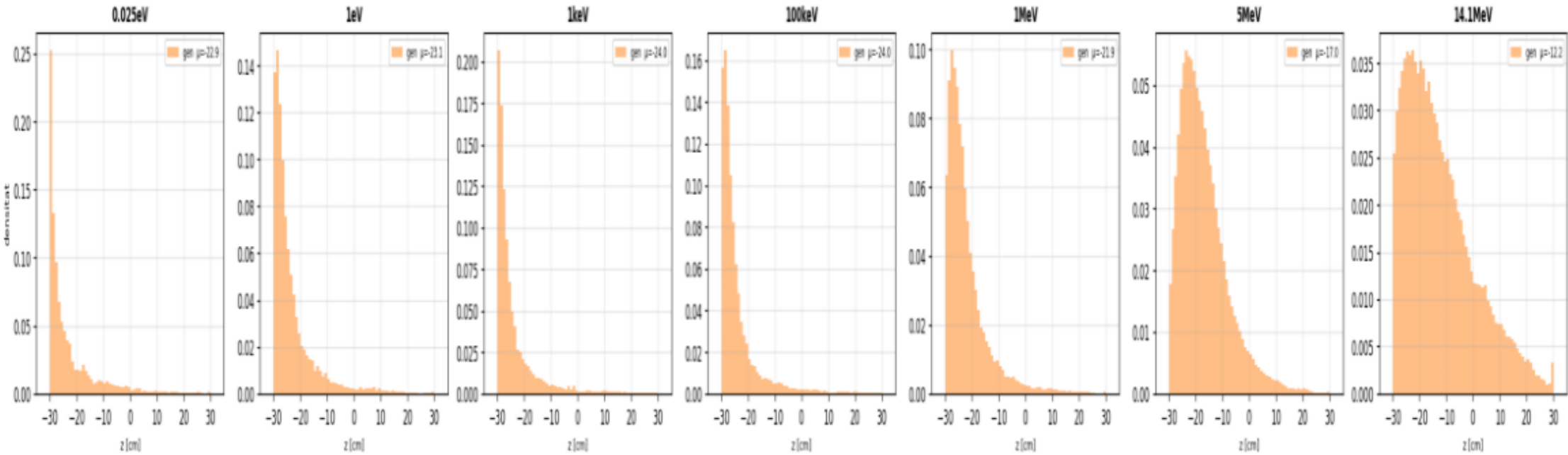
Dataset
(shared)

Paràmetre	Valor
z_mean	-18.052
z_std	10.603
R_max	20
log_edep_mean	-4.4352
log_edep_std	2.9554
edep_offset_MeV	1e-12
log_t_mean	3.6413
log_t_std	1.8725
t_offset_ns	0.001
log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

C z phys

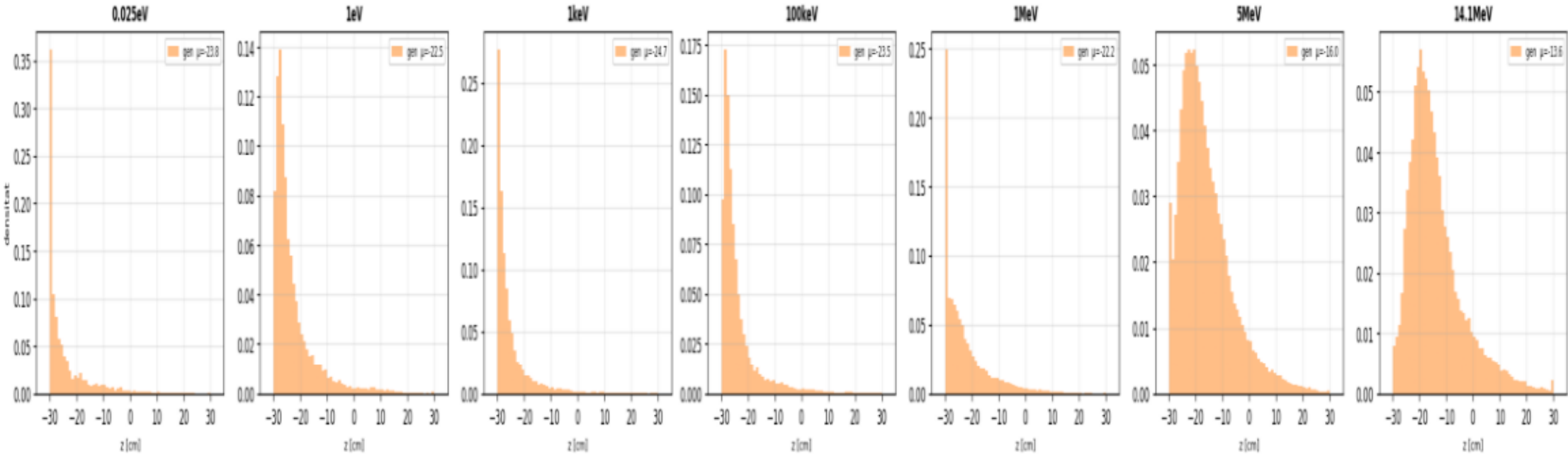
Part C – Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPiC-FM)

run_013 (Linear)



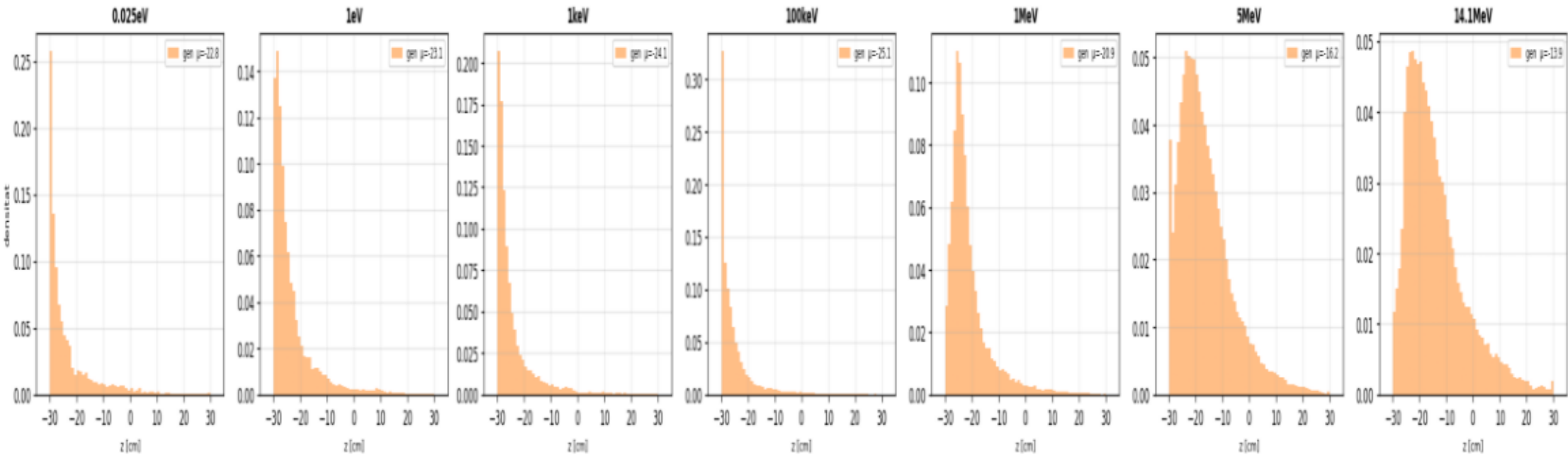
Part C – Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPiC-FM)

run_014 (bins=8)

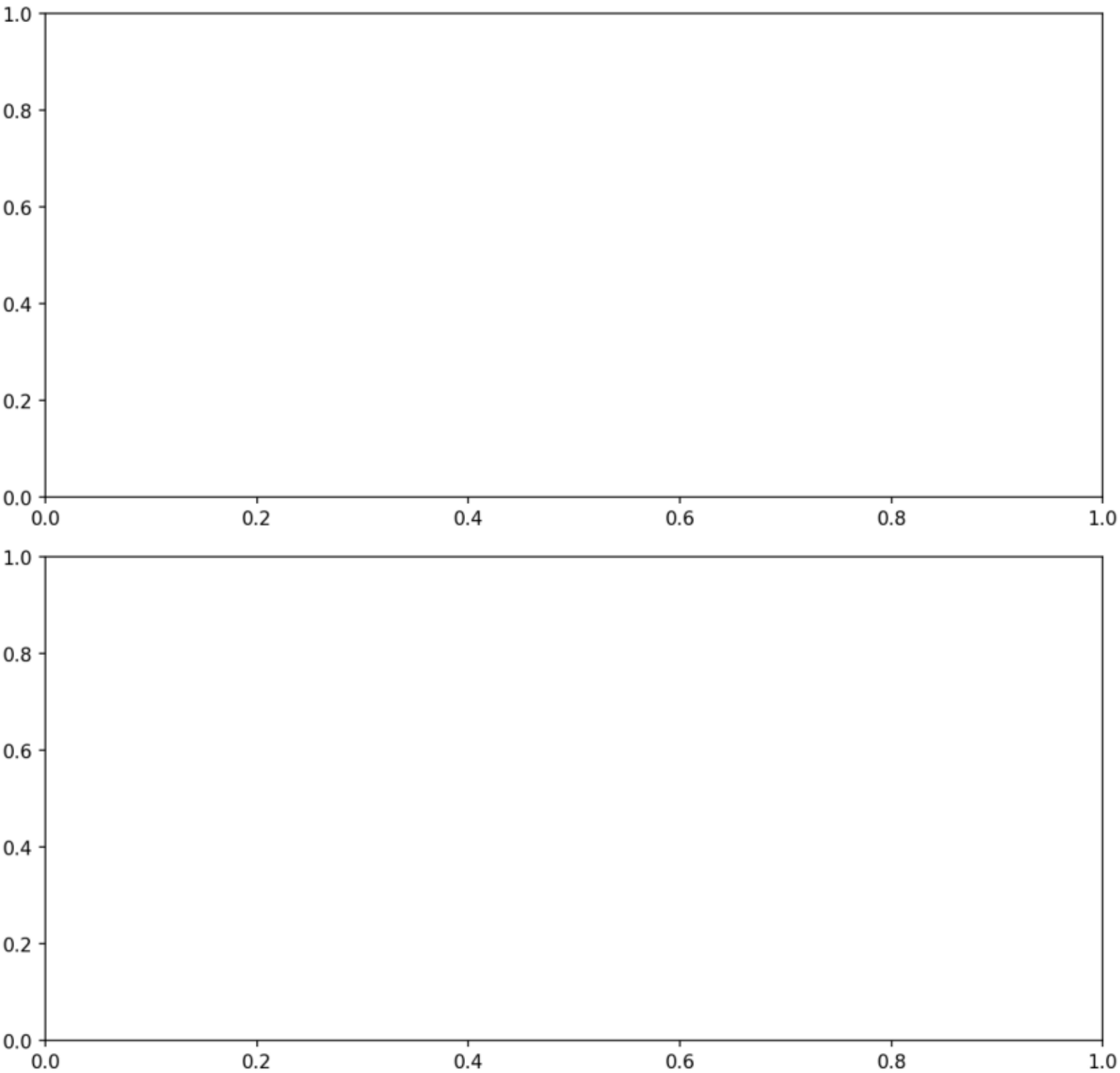


Part C – Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPiC-FM)

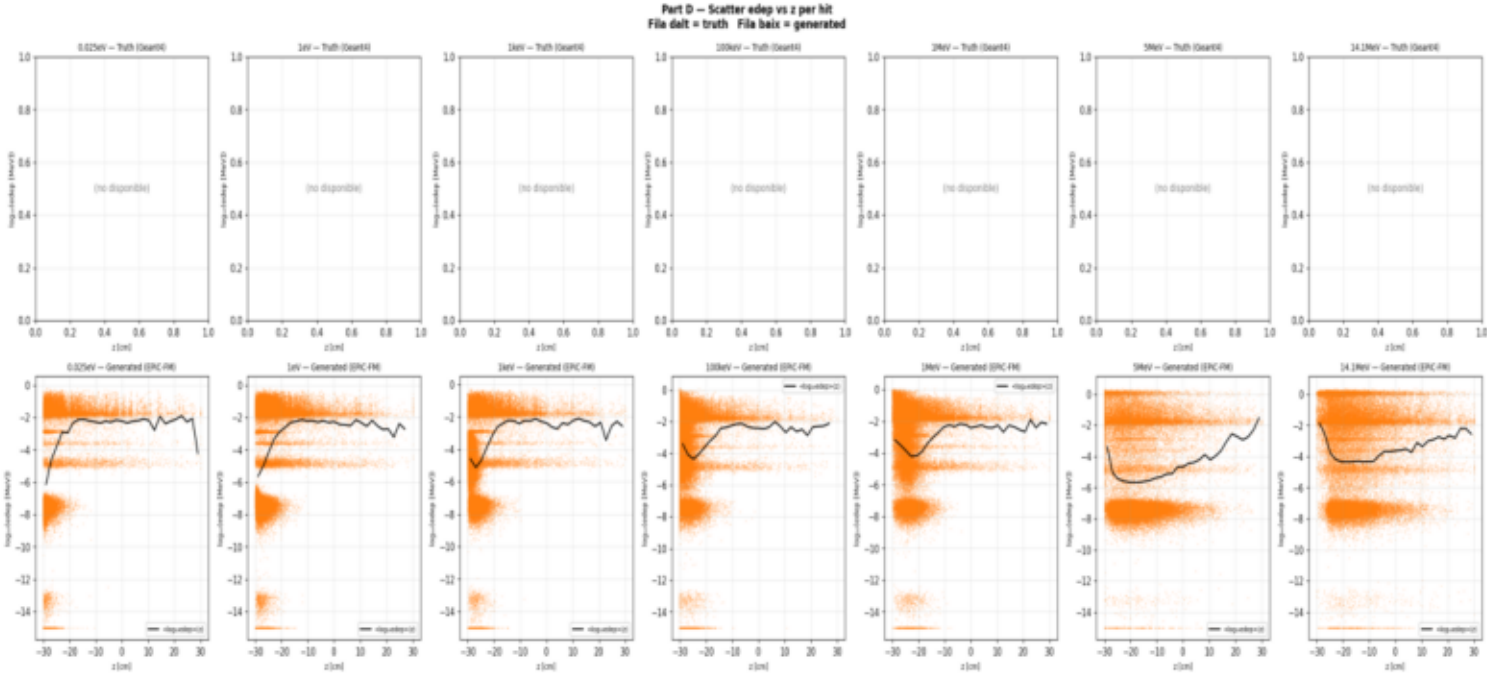
run_015 (bins=16)



D edep vs z scatter



run_015 (bins=16)



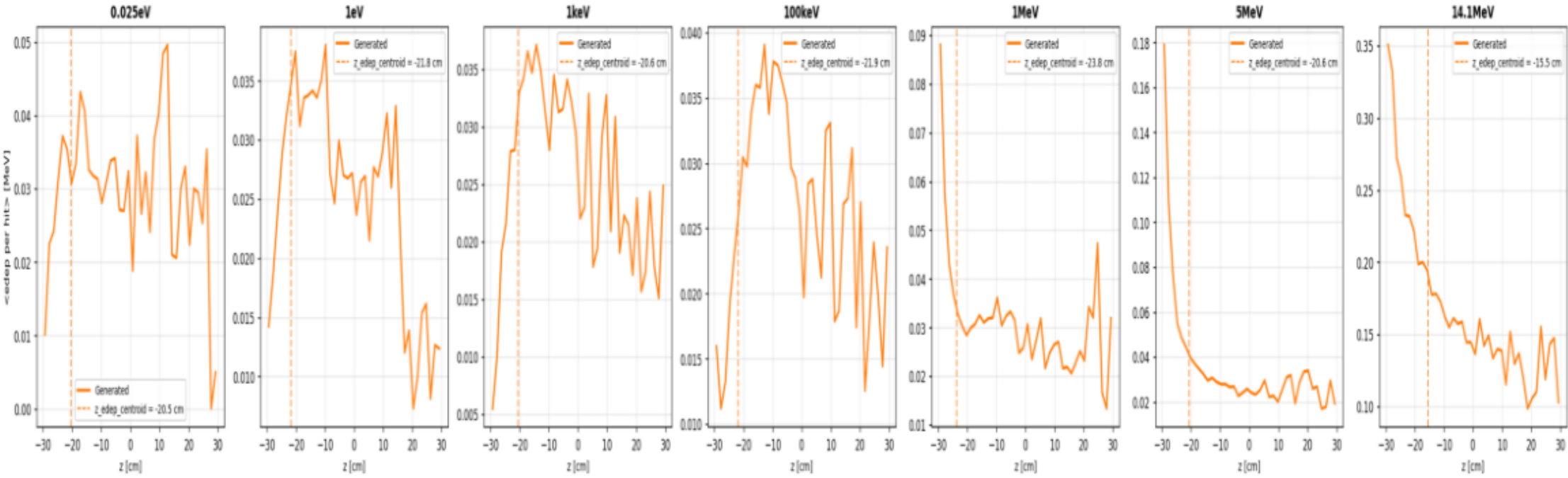
Bandes horizontals esperades (neutrons en CH2):

log10(edep) ~ +0.65	12C(n,n'g) inelastic, g = 4.44 MeV
log10(edep) ~ +0.35	H(n,g)D captura termica H, g = 2.22 MeV
log10(edep) ~ +0.10	12C(n,g) captura C, g = 1.26 MeV
continu	n(H,H)n recul de proto (elastic), E in [0, En]
continu	Compton / fotoelectric de g secundaris
log10(edep) ~ -14	Floor de tracking Geant4 (electrons baixa E)

E edep z profile

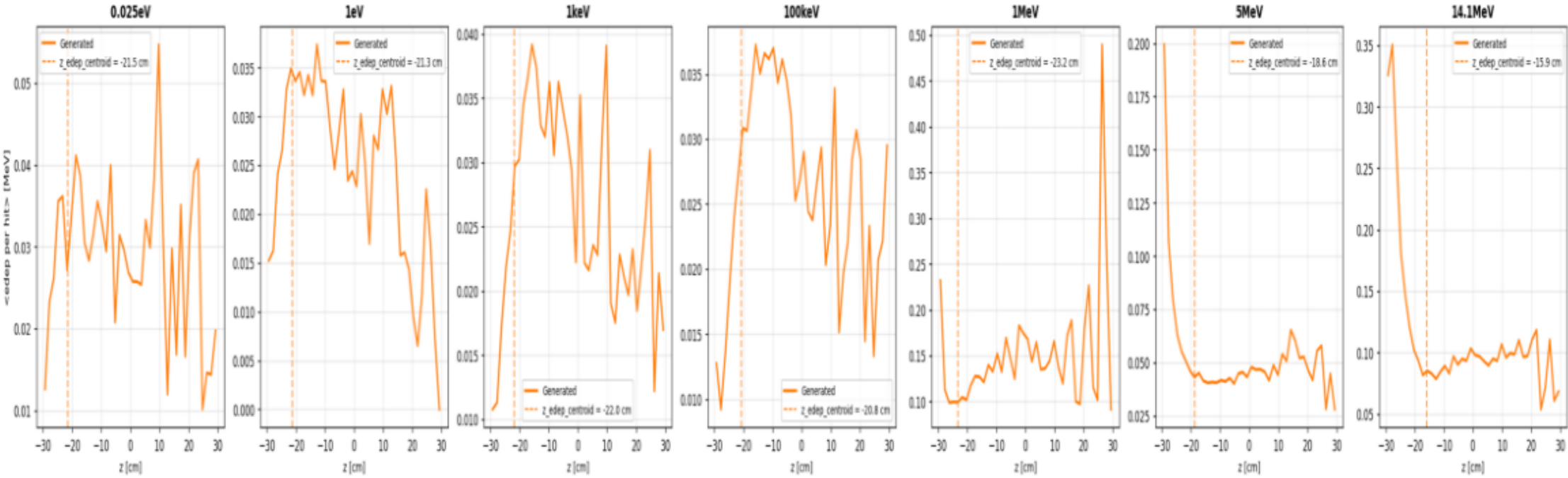
Part E – Perfil <edep>(z) binned
Blau = truth Taronja = generated (les línies verticals = centroide z)

run_013 (Linear)



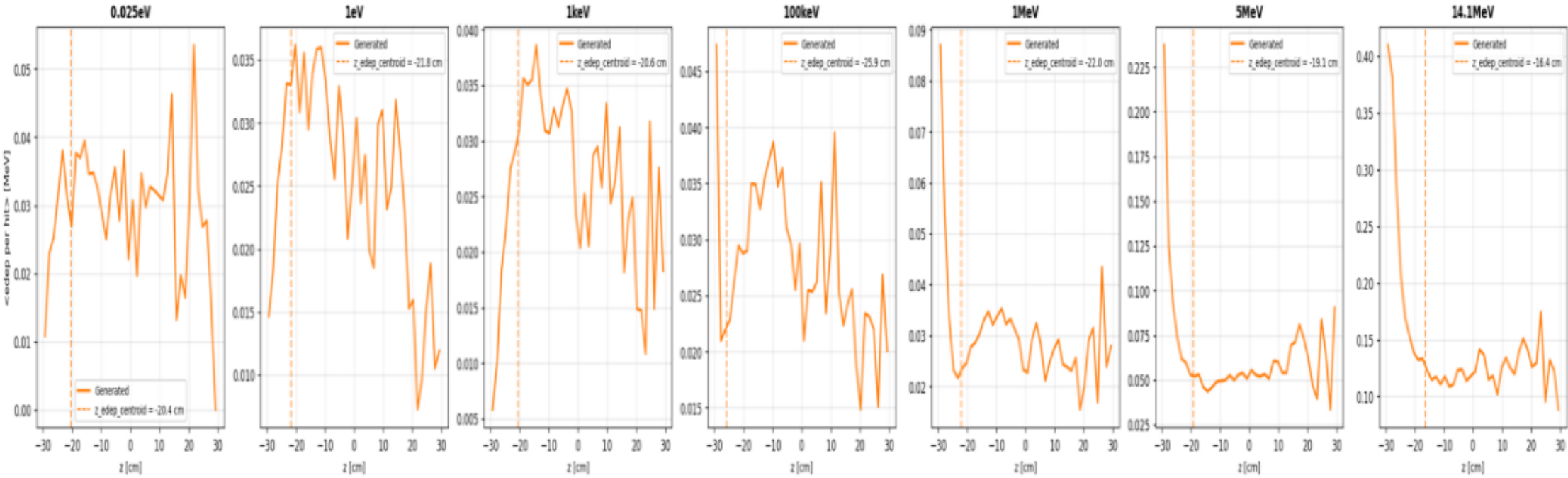
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run_014 (bins=8)

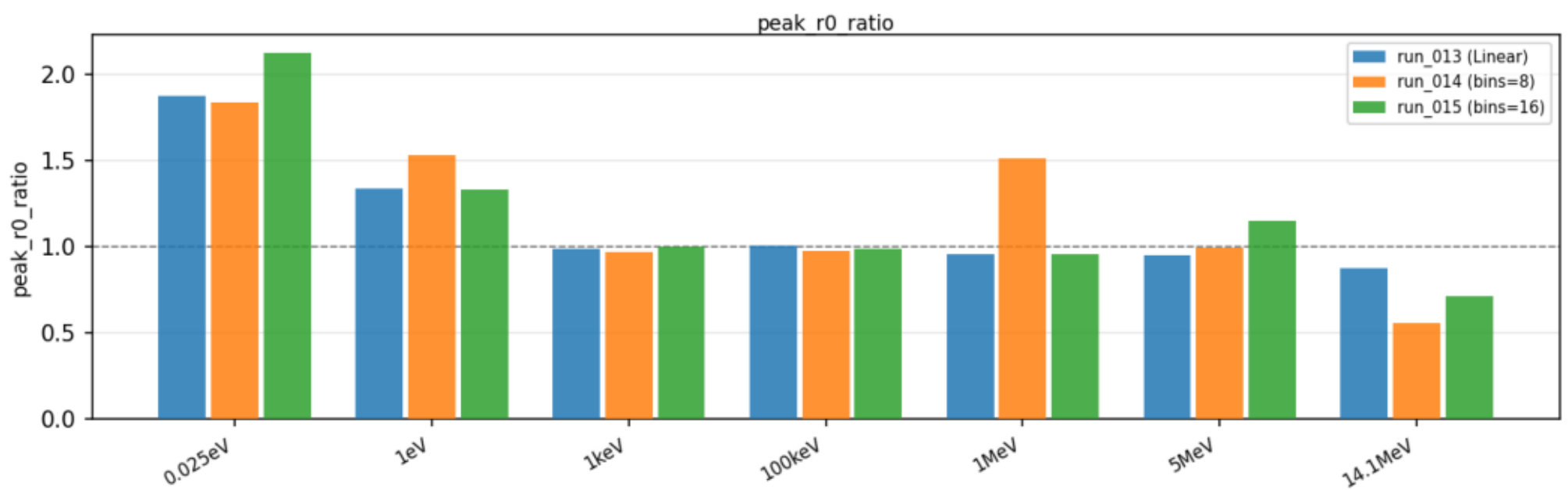
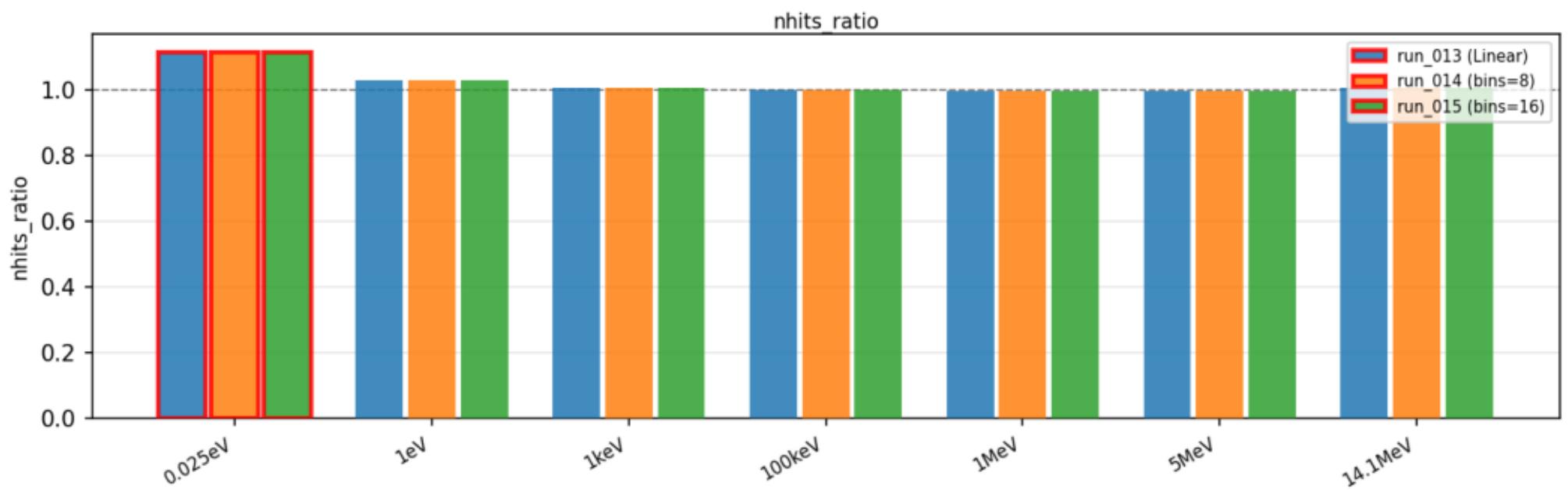
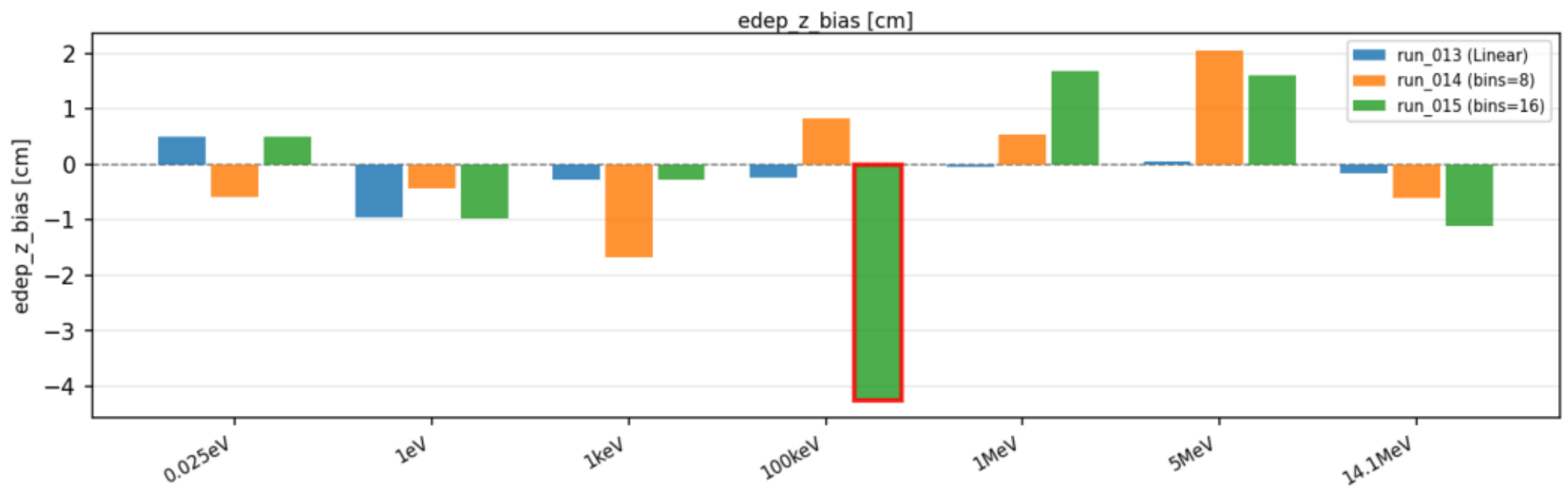
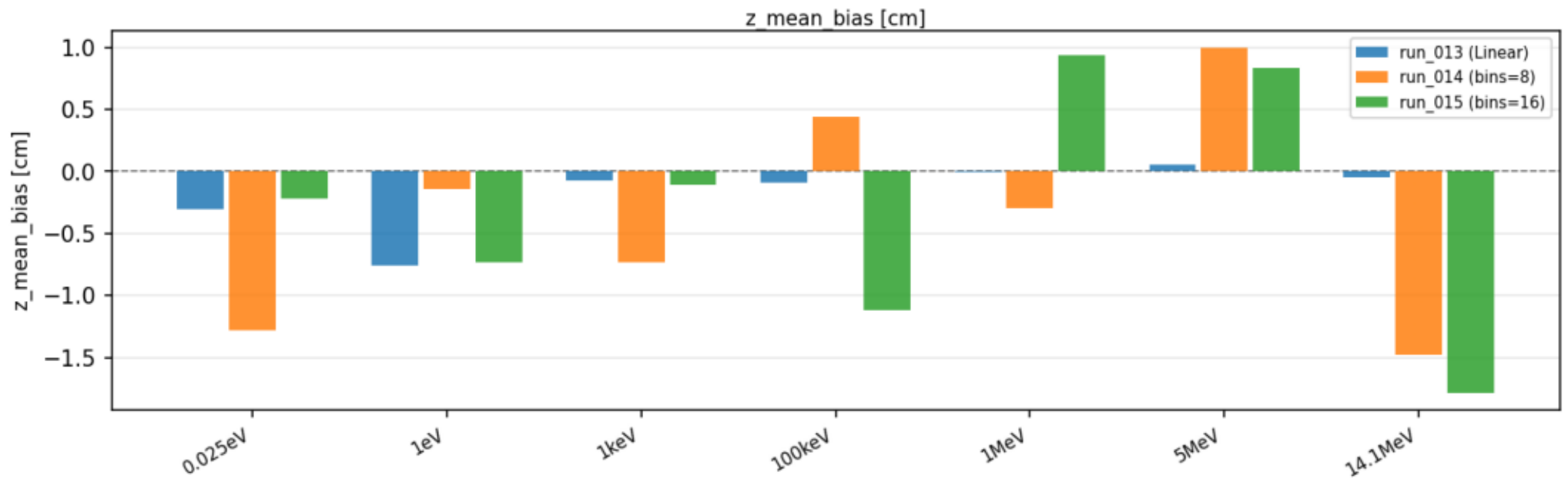


Part E – Perfil <edep>(z) binned
Blau = truth Taronja = generated (les línies verticals = centroide z)

run_015 (bins=16)



Part M - Resum metriques per run i energia



Part Z - Heatmap metriques OK/WRN/BAD per run i energia (real RI, gen vs truth)

